

Evaluating the Evaluations: A Perspective on Benchmarks

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Abstract

More and more benchmarks, datasets, and evaluation tasks are becoming available. This is extremely useful for the community because it enables researchers and practitioners to test and evaluate new techniques. However, the construction, evaluation, and maintenance of data sets and benchmarks is opaque which creates problems with respect to stability and true representations. Our position is that we need to revisit how we design and implement benchmarks. The SPEC benchmark offers interesting perspectives that our community should consider. We use a data set of influential papers and resources to discuss important benchmark aspects such as realistic workloads, reliability, validity, leakage, and labeling. We conclude by proposing a list of principles for constructing evaluation benchmarks.

1 Introduction

Evaluation and benchmarking are at the center of IR, NLP, ML, and related areas. Quantifying improvements of new methods is very important for measuring progress. There are a huge number of datasets and benchmarks available online. There will surely be even more such resources in the future because these resources are extremely useful for the community; they have made it possible for researchers and practitioners to test and evaluate new techniques. Testing and evaluating the quality of a model can be done in a few steps and provide signal on next steps. A problem with this black-box approach is dependence on assumptions that are often unstated, involving data quality, reliability and validity.

Because these resources play such a crucial role, we think that there should be more in-depth assessment and peer reviewing, not only of methods but also of benchmarks. This also applies in industrial settings where domain specific evaluations are conducted. The construction, evaluation, and maintenance of such benchmarks should not only be more transparent but should also be held to software engineering standards.

The data collection, validation, and curation processes of different benchmarks and evaluation data sets raise a number of methodological concerns. Business decisions based on missing

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data points, incorrect values, stale values and misleading values can lead to disastrous customer experience and low adoption.

Experiments are an important part of the research process in our community and one may argue that it is very difficult to get published without them. Given the importance of experiments and benchmarks in our field, we're interested in evaluating the evaluations.

To improve reliability, a benchmark should be structured to enable users to measure agreement of labels from different sources. A common problem with validity is leakage. If there is leakage between the training and test sets, it is possible to do well on the test without mastering the material. The tendency to increase the number of tasks makes it difficult to ensure high quality in terms of validity and reliability. To ensure transparency, a benchmark should share necessary details, especially on data collection and label assignment.

Tasks should be relevant to use cases. There should be more characterizations of use cases and workloads in our field. A benchmark can be misleading if the sample is not representative of a population of interest. A benchmark needs to be used as a learning tool for researchers and practitioners and not only as leader board.

We start by listing influential papers and resources, organized by dimensions of success. As part of our data analysis, we identify issues and challenges. As an example, from a design perspective, the TREC QA benchmark seems more technically solid compared to SQuAD but it has fewer citations. We conclude the paper with a list of principles that are especially important for high-stakes use cases with commercial relevance.

2 Analysis of Influential Articles

Consider the following dimensions of success:

1. **Awards:** Test of Time
2. **Popularity Metrics:** Citations in Google Scholar and Semantic Scholar [Wade, 2022], likes on HuggingFace (HF),¹ stars on Papers with Code (PwC)²
3. **Interdisciplinary Impact:** Benchmarks that appeal to more fields and larger fields tend to be more influential (in terms of popularity metrics) than those that appeal to fewer fields.
4. **Technical Merit:** Reliability, Validity, etc.
5. **Commercial Impact:** See Section 3.2.

We will discuss three tables that list a number of influential papers along with popularity metrics such as citations in Semantic Scholar:

- Table 1 lists some influential TREC papers sorted by popularity.
- Table 2 shows papers that won Test of Time awards up to 2017 [Allan et al., 2017], sorted by time.
- Table 3 lists some influential resources, benchmarks and models in IR and related fields.

Before discussing these tables in more detail, we are concerned that benchmarks have a way of taking on a life of their own, as will be discussed in Section 5 on the Penn TreeBank (PTB).

¹<https://huggingface.co/>

²<https://paperswithcode.com/>

Cites	Reference	Title
2360	[Robertson et al., 1994]	Okapi at TREC-3
1138	[Voorhees, 1999]	The TREC-8 Question Answering Track Report
1071	[González and Gómez, 2007]	TREC: Experiment and evaluation in information retrieval
707	[Voorhees, 2004]	Overview of the TREC 2003 Question Answering Track
629	[Buckley et al., 1994]	Automatic Query Expansion Using SMART: TREC 3
459	[Garofolo et al., 2000]	The TREC Spoken Document Retrieval Track...
457	[Hersh et al., 2007]	TREC 2007 Genomics Track Overview
406	[Clarke et al., 2009]	Overview of the TREC 2009 Web Track
397	[Robertson et al., 1995]	Okapi at TREC-4
393	[Craswell et al., 2020]	Overview of the TREC 2019 deep learning track
393	[Voorhees and Tice, 2000]	The TREC-8 Question Answering Track Evaluation
385	[Craswell et al., 2005]	Overview of the TREC 2005 Enterprise Track
329	[Craswell et al., 2021]	Overview of the TREC 2020 Deep Learning Track
308	[Jaleel et al., 2004]	UMass at TREC 2004: Novelty and HARD
245	[Ounis et al., 2011]	Overview of the TREC 2011 Microblog Track
179	[Roberts et al., 2014]	Overview of the TREC 2014 Clinical Decision Support Track
177	[Voorhees and Hersh, 2012]	Overview of the TREC 2012 Medical Records Track
173	[Alonso and Mizzaro, 2012]	Using crowdsourcing for TREC relevance assessment

Table 1. Citations in Semantic Scholar for some influential TREC papers.

The authors, [Marcus et al. \[1993\]](#), originally designed the PTB for one purpose (parsing), but it has been recently repurposed by the community for a very different use (perplexity) in a way that is not as meaningful as it might appear. Once a benchmark is released into the wild, it has a tendency to take on a life of its own. We, as a community, should be more careful with credit assignment, especially when introducing serious weaknesses into other people’s work.

2.1 Benchmarks Should Make a Point

We view benchmark design like Brooks views programming languages. Brooks managed the development of software for IBM mainframe computers and later wrote candidly about that experience in *The Mythical Man-Month* [[Brooks, 1995](#)]. The second author heard Brooks give a talk at Bell Labs in front of the people responsible for Unix. This talk referred to Figure 1 of [[Brooks, 1987](#)]:

- More exciting products: Unix, APL, Pascal, Modula, Smalltalk, Fortran
- Less exciting products: Cobol, PL/1, Algol, MVS/370, MS-DOS

where Brooks asserted that all of these products were successful, at least in terms of awareness. His point was that the more exciting products were designed by a few remarkable people making a point; the less exciting products were designed by committees making compromises:

The differences are not minor—it is rather like Salieri and Mozart. Study after study shows that the very best designers produce structures that are faster, smaller, simpler, cleaner, and produced with less effort. The differences between the great and the average approach an order of magnitude. [[Brooks, 1987](#)]

Reference	Cites	Title
[Katzner et al., 1983]	193	A study of the overlap among document representations
[Voorhees, 1985]	189	The cluster hypothesis revisited
[Rijsbergen, 1986]	215	A New Theoretical Framework for Information Retrieval
[Fagan, 1987]	40	Automatic phrase indexing for document retrieval
[Jones, 1988]	24	A Look Back and a Look Forward
[Harman, 1988]	13	Towards interactive query expansion
[Furnas et al., 1988]	98	Information retrieval using a singular value decomposition model of latent semantic structure
[Belew, 1989]	99	Adaptive information retrieval: Using a connectionist representation to retrieve and learn about documents
[Pejtersen, 1989]	59	A library system for information retrieval based on a cognitive task analysis and supported by an icon-based interface
[Turtle and Croft, 1989]	454	Inference networks for document retrieval
[Cutting et al., 1992]	1830	Scatter/Gather: a cluster-based approach to browsing large document collections
[Callan et al., 1995]	665	Searching distributed collections with inference networks
[Xu and Croft, 1996]	1298	Query expansion using local and global document analysis
[Singhal et al., 1996]	612	Pivoted Document Length Normalization
[Allan et al., 1998]	621	On-line new event detection and tracking
[Bharat and Henzinger, 1998]	736	Improved algorithms for topic distillation in a hyper-linked environment
[Ponte and Croft, 1998]	695	A language modeling approach to information retrieval
[Carbonell and Goldstein-Stewart, 1998]	2014	The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries
[Hofmann, 1999]	2094	Probabilistic Latent Semantic Indexing
[Berger and Lafferty, 1999]	740	Information retrieval as statistical translation
[Herlocker et al., 1999]	2453	An algorithmic framework for performing collaborative filtering
[Buckley and Voorhees, 2000]	205	Evaluating Evaluation Measure Stability
[Järvelin and Kekäläinen, 2000]	1356	IR evaluation methods for retrieving highly relevant documents
[Lafferty and Zhai, 2001]	198	Document language models, query models, and risk minimization for information retrieval
[Lavrenko and Croft, 2001]	1742	Relevance-Based Language Models
[Zhai and Lafferty, 2001]	1321	A study of smoothing methods for language models applied to Ad Hoc information retrieval

Table 2. Award Winning Papers: Test of Time (from [Allan et al., 2017]).

Resource/Benchmark/Model	Semantic Scholar	Google Scholar	HF Likes	PwC Stars
ResNet [He et al., 2016]	177,065	244,794	739	77,201
BERT [Devlin et al., 2019]	84,392	119,203	1.91k	135,247
ImageNet [Russakovsky et al., 2015]	37,332	47,929	416	746
GPT-3 [Brown et al., 2020]	33,349	35,576	645	
CIFAR [Krizhevsky et al., 2009]	32,366	31,907	62	
GPT-2 [Radford et al., 2019]	19,527	14,289	2.37k	
Penn TreeBank (PTB) [Marcus et al., 1993]	8802	10,854	15	
SQuAD [Rajpurkar et al., 2016]	7492	8903	265	1403
GLUE [Wang et al., 2018]	6415	7611	372	1647
LibriSpeech [Panayotov et al., 2015]	5791	7130	129	
cc100 [Conneau et al., 2020]	5743	6372	74	
Visual Genome [Krishna et al., 2017]	5258	6043	66	256
PaLM [Chowdhery et al., 2022]	5168	5022		
IMDB [Maas et al., 2011]	4894	6336	250	350
Reuters-21578 [Lewis et al., 2004]	2997	3784	16	
Natural Questions [Kwiatkowski et al., 2019]	2699	2805	85	
SQuAD 2.0 [Rajpurkar et al., 2018]	2591	3063	179	
MS Marco [Bajaj et al., 2016]	2394	786	123	
Wikitext-103 [Merity et al., 2016]	2346	2408	6	
Switchboard [Godfrey et al., 1992]	2245	2936	111	
MNIST [LeCun et al., 2010; Deng, 2012]	2163	5750	113	
SuperGLUE [Wang et al., 2019]	2089	2309	156	1647
The Pile [Gao et al., 2020]	1682	645	399	
MUSE [Conneau et al., 2017]	1587	1328		3190
CogComp/trec [Li and Roth, 2002]	1529	1907	40	
BIG-Bench [Srivastava et al., 2022]	1432	1146	61	
Common Voice [Ardila et al., 2020]	1334	1636	195	
Brown Corpus [Francis and Kučera, 1982]	1262			
TIMIT [Garofolo et al., 1983]	1067	3832	21	
FLORES [Goyal et al., 2022]	462	417	66	
ERNIE [Sun et al., 2021]	356	463	92	11,655
MRQA [Fisch et al., 2019]	285	295	20	292
MUC-3/4 [Sundheim, 1991, 1992]	69/129	94/155		
CulturaX [Nguyen et al., 2023]	60	61	475	
allenai/c4			314	
DuReader [He et al., 2018]	12	316	10	6918
Callhome [Canavan et al., 1997]		223	13	
WMT-19 [Wikimedia, 2019]		28		

Table 3. Metrics for some influential resources: HF = HuggingFace; PwC = Papers with Code. Note that SQuAD (7492 for [Rajpurkar et al., 2016]) has more citations than TREC-8 (1138 for [Voorhees, 1999]), though TREC-8 is closer to real-world workloads, as discussed in Section 3.3.

Committees do what committees do. Bloat is inevitable. Committees are good for reaching consensus, but not for creativity and inspiration. BIG-bench is an example of a committee effort:

BIG-bench currently consists of 204 tasks, contributed by 450 authors across 132 institutions. [Srivastava et al., 2022]

The Mythical Man-Month is deeply skeptical of large teams. It is hard to make a point with hundreds of authors. A benchmark should have a single point (and not too many points); it should be large enough to make that point, and no larger.

SPEC³ [Dixit, 1993] has had more commercial impact than benchmarks in our field because it focuses on a question that matters to an important audience (suppliers and consumers of CPUs): *which CPU should I buy?* Like many benchmarks in our field, SPEC is a collection of programs (tasks), intended to measure different needs of different users. Some programs are bounded by cycles and some are bounded by memory.

In *Summarizing performance is no mean feat*, Mashey [2005] argues that geometric means are better than arithmetic means, as a single metric for different customers with different workloads. So too, our benchmarks should be designed to address different workloads. Elsewhere,⁴ Mashey suggests replacing arithmetic means in our benchmarks with geometric means.⁵ It is hard to interpret averages over tasks using metrics with incompatible units: accuracy, correlations, F1, etc. If we report performance relative to a baseline, then the units are dimensionless, and can be combined with geometric means.

In this way, SPEC makes an important point, establishing that modern CPUs are 50,000 times faster than they were 40 years ago (relative to a VAX 11/780 in 1978).⁶ In [Church and Alonso, 2024], we suggest following Mashey’s advice so many of our benchmarks can be used to make a similar point. The tech sector has benefited by its ability to demonstrate remarkable progress over decades. So too, we should be able to use our benchmarks to make statements about performance of LLMs over years/decades relative to a baseline such as human performance. Too many of our benchmarks are used for (pointless) SOTA-chasing [Church and Kordoni, 2022].

3 What Has Worked Well in the Past?

This section will discuss Tables 1-3, as well as measures of success: awards (Test of Time), popularity metrics (citations), interdisciplinary impact, technical merit and commercial impact.

3.1 Popularity Metrics and Interdisciplinary Impact

Tables 1, 2 & 3 show influential papers with popularity metrics such as citations in Semantic Scholar. Table 2 (Test of Time) is sorted by time, unlike Tables 1 & 3, which are sorted by

³<https://www.spec.org/>

⁴https://github.com/kwchurch/Benchmarking_past_present_future?tab=readme-ov-file#Mashey

⁵Robertson [2006] suggested using geometric mean to aggregate over items in a test set.

⁶See slide 9 of <https://courses.grainger.illinois.edu/CS433/fa2021/slides/chapter1-part1-post-lecture.pdf>; there is a similar chart in Figure 1.1 of [Hennessy and Patterson, 2012].

citations. Citations tend to increase with recency because the literature is expanding rapidly, doubling every decade or two⁷ [Price, 1965; Bornmann and Mutz, 2014; Bornmann et al., 2021].

It is useful to compare Table 1, influential papers in TREC [Voorhees and Harman, 2005], with Table 3, influential resources in related fields. In addition to recency, citations also increase for resources that appeal to more fields and larger fields. For example, tasks such as question answering have been studied in IR (TREC-8) as well as ML and NLP (SQuAD), though SQuAD has more citations than TREC-8 (7492 versus 1138), perhaps because SQuAD is more recent and it appeals to more fields and larger fields.

Popularity metrics such as citations are a good place to start because they are easy to collect, though there are a number of drawbacks. In particular, although SQuAD may have more citations than TREC-8, we are not convinced that SQuAD is better, as will be discussed below in sections on technical merit and commercial impact.

SOTA-chasing is another drawback. SOTA-chasing attracts even more citations than interdisciplinary appeal, despite the fact that SOTA-chasing encourages pointless activities that are unlikely to generalize to commercially important use cases. TREC has resisted the temptation to promote SOTA-chasing, emphasizing cooperation over competition:⁸

*competing may give you a bigger piece of the pie ...
... while cooperation makes the whole pie bigger*⁹

TREC participants are asked to sign a form that forbids explicit advertising of TREC results. This prohibition was mentioned a number of times in the videos of the 25th anniversary of TREC.¹⁰ While participants appreciate the principle, the temptation to boast is difficult to resist. Since SOTA-chasing is popular, TREC’s rules against SOTA-chasing may have consequences in terms of popularity metrics.

3.2 Technical Merit and Constructed Queries

Setting popularity metrics aside for the moment, based on technical merit, how do the benchmarks in Table 3 compare with the the benchmarks in Table 1? When evaluating benchmarks, we should be concerned about classic questions in psychology such as reliability and validity [Krippendorff, 2018], which will be discussed in more detail in Section 4.

More specifically, how does SQuAD compare with TREC-8? One difference involves the methods used to generate questions:

- SQuAD (crowdsourcing): popular in academia (especially when logs are unavailable)
- TREC-8 (logs): more common in industry (especially when logs are available)

⁷<https://blogs.nature.com/news/2014/05/global-scientific-output-doubles-every-nine-years.html>

⁸https://github.com/kwchurch/Benchmarking_past_present_future/blob/master/README.md#Voorhees

⁹slides 6-7 of https://github.com/kwchurch/Benchmarking_past_present_future/blob/master/slides/session3/ElLEN_benchmarking.pdf

¹⁰<https://trec.nist.gov/celebration/25thcelebration.html>

Here are some examples of questions in TREC-8 and SQuAD:

- TREC-8 questions (from Figure 1 of [Voorhees, 1999]):
 1. *How many calories are there in a big mac?*
 2. *What two US biochemist won the Nobel Prize in medicine in 1992?*
 3. *Who was the first American in space?*
- A SQuAD question with context and gold answer:
 - **Question:** *What year was the DOE report about fuel consumption published?*
 - **Context:** *Several studies have suggested that DST increases motor fuel consumption. The 2008 DOE report found no significant increase in motor gasoline consumption due to the 2007 United States extension of DST.*
 - **Answer:** *2008* (starts at character offset 78 in context).

The QA track in TREC started with constructed questions in 1999, but moved to query logs for subsequent TREC QA tracks (2000-2007) because constructed questions are too easy for systems, and unrealistic [Voorhees, 2001]. TREC-8 questions were extracted from logs of a system called FAQFinder [Hammond et al., 1995]. These questions were submitted by undergraduate students “who were genuinely interested in the answers to the questions.”¹¹

SQuAD starts with a context from a Wikipedia article such as this¹² and then MTurk workers generate questions based on these contexts. The gold answers are substrings of the context. Note that SQuAD questions often depend on the context, as in this case. When we give the SQuAD question above to search engines in today’s context, they tell us about more recent DOE reports.

Crowdsourcing is a popular strategy for collecting data, especially when you do not have big tech resources. It was used for 9 of 18 question-answering benchmarks in MRQA (See Table 1 in [Fisch et al., 2019]):

- Crowdsourced (9 benchmarks): ComplexWebQ, DROP, DuoRC, HotpotQA, MCTest, NewsQA, QAMR, SQuAD, TREC
- Domain Experts (5 benchmarks): BioASQ, BioProcess, QAST, RACE, TextbookQA
- Search logs (1 benchmark): Natural Questions
- Jeopardy (1 benchmark): SearchQA
- Synthetic (1 benchmark): RelationExtraction
- Trivia (1 benchmark): TriviaQA

Industry is more likely to have access to search logs and other instrumentation. Natural Questions [Kwiatkowski et al., 2019] is the only benchmark in MRQA to use search logs. Most of the authors of benchmarks work in academia and do not have access to logs, but Natural Questions is a notable exception.

¹¹See Section 2.1 of [Voorhees and Tice, 2000]

¹²https://en.wikipedia.org/wiki/Analysis_of_daylight_saving_time

3.3 Commercial Impact: Lessons from SPEC

Since TREC is organized by NIST, which is part of the US Department of Commerce, it may not be surprising that benchmarks from TREC would be more closely related to commercial concerns than many of the benchmarks in Table 3. We ran Microsoft Bing and Google on questions in the TREC-8 and SQuAD examples above. Instant answers are more successful for TREC-8 questions than SQuAD questions, suggesting search companies place a higher priority on TREC-8 questions. TREC-8 may come closer to real-world workloads.

In terms of commercial impact, SPEC [Dixit, 1993] has had more influence than most benchmarks in Tables 1-3 because SPEC addresses a question with more commercial importance: *which computer should I buy?* Customers want a single number, not a long complicated story such as a precision/recall curve. Consumers will not read a thoughtful review like you might see in Consumer Reports. A simple credible number is also good for suppliers. Previous benchmarks such as Whetstone¹³ and Dhrystone¹⁴ encouraged pointless SOTA-chasing exercises that do not generalize to commercial use cases.

4 Realistic Use Cases and Workloads

Typical workloads are a moving target in our community. Over the past year or so, usage of AI-based applications has increased considerably. As workloads evolve, so too, benchmarks need to catch up. Ideally, performance on benchmarks should be a leading indicator of what customers will experience, not a lagging indicator. One of the principles in Section 9 involves maintenance and evolution.

The challenge is to come up with a single number that is useful for predicting performance over a wide variety of workloads. SPEC replaced previous benchmarks such as Whetstone and Dhrystone with a larger set of larger programs, intended to be more representative of typical workloads. The older benchmarks tended to focus on cycles – integer and floating point arithmetic (MIPS and FLOPS); SPEC added concerns that came along later such as workloads that made use of vector arithmetic and memory. It is standard practice in many fields to estimate workloads: databases [Zhang et al., 2018], networking [Calzarossa et al., 2016], and web search [Broder, 2002].

We are facing similar challenges with benchmarks for IR and LLMs. We would like to come up with a single number that generalizes over many (unknown) use cases. One might hope that averaging over many tasks might address this concern, though SPEC was designed to meet the needs of a clearly identified community. For benchmarks such as GLUE, it is harder to characterize (a) typical workloads, (b) user communities and (c) their use cases.

There are a few benchmarks in our field that are based on data from real customers [Kwiatkowski et al., 2019], but most benchmarks in our field were not designed to be representative of typical workloads. SPEC is different from many of the academic benchmarks in our field because many computer companies chose to use SPEC, whereas in our field, companies have their own internal benchmarks that are more representative of the needs of their customers. Going forward, when we build new benchmarks, we should aim to come up with a benchmark like SPEC that is so

¹³<https://www.netlib.org/benchmark/whetstone.c>

¹⁴<https://en.wikipedia.org/wiki/Dhrystone>

compelling that industry would choose to use our benchmark instead of building their own. When we list principles in Section 9, a key metric of success will be adoption.

There is recent work on search evaluation that suggests moving away from evaluating ranking quality to a more utilitarian approach [Broder and McAfee, 2023]. The authors argue that algorithmic ranking partially determines search utility and that there are search engine aspects (or features) that help users with their tasks. Thus, we think that a realistic workload should be as close as possible to user goals. There should be fewer tasks that are more relevant, as opposed to many irrelevant tasks, found in many benchmarks.

5 Benchmark Deep-Dive into the Penn TreeBank (PTB)

This section will do a deep dive into the Penn TreeBank (PTB) [Marcus et al., 1993], a benchmark that is often used for two very different purposes. The authors of this benchmark originally designed the PTB for one purpose but it has been recently repurposed by the community for something else in ways that are not as meaningful as it might seem. It is not clear who did this when and why, but now that the mistake has become standard practice, reviewers are likely to reject papers that do not repeat the mistake. More generally, we are concerned about how current practice is using benchmarks in ways that can be hard to defend, and diverge considerably from the original intention of the authors.

Here are some highly cited papers that use the PTB in these two ways:

- Parsing: [Collins, 1997, 2003; Charniak, 2000]
- Perplexity: [Zhao et al., 2023; Zhuang et al., 2020; Dong and Yang, 2019; Khandelwal et al., 2018]

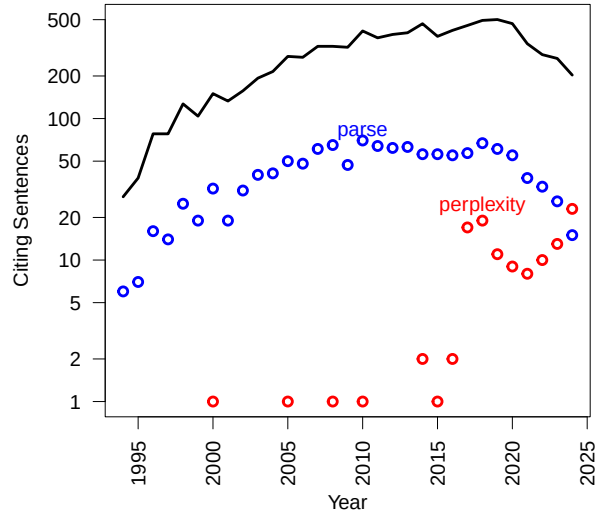
Figure 1a uses citing sentences to show how the community views the PTB. The parsing usage used to be more popular, but recently, the perplexity use has been gaining popularity. The black line in Figure 1a shows the distribution of 8677 papers that cite the PTB over time. The blue points counts papers that cite PTB using the word “parse” in the citing sentence. The red points are similar to the blue points but for the word “perplexity.” Note that the blue points peaked several years ago, whereas the red points are larger in more recent years, indicating that the use of the PTB for perplexity is more recent than the use of the PTB for parsing.

5.1 Inappropriate Processing

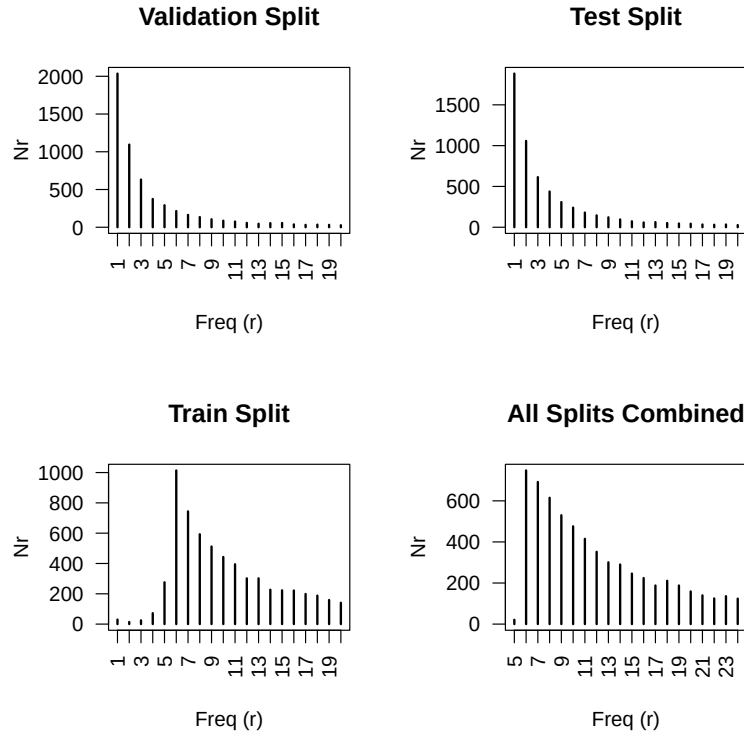
What is wrong with the text-only version¹⁵ of PTB? After rare words and numbers were removed by the community downstream of the LDC version,¹⁶ <unk> (rare words) has become the most common “word” in the test set, as shown in Table 2a. We should not confuse <unk>-prediction with perplexity; perplexity has many important applications [Shannon, 1948, 1951] unlike <unk>-prediction. Note that estimates of perplexity becomes trivial (and completely meaningless) if we replace all words with <unk>; the more words we replace with <unk>, the easier (and less

¹⁵<https://huggingface.co/ptb-text-only>

¹⁶<https://catalog.ldc.upenn.edu/LDC99T42>



(a) Black line shows 8677 citing sentences that refer to PTB. Blue shows citing sentences that mention *parse*. Red shows citing sentences that mention *perplexity*.



(b) N_r is the number of words that appear r times. We expect $N_r \gg N_{r+1}$, but the bottom row has surprisingly small N_r for $r \leq 5$, because of Figure 2.

Figure 1. Penn TreeBank (PTB) [Marcus et al., 1993] was originally constructed to study parsing, though more recently, it has been repurposed to also study perplexity.

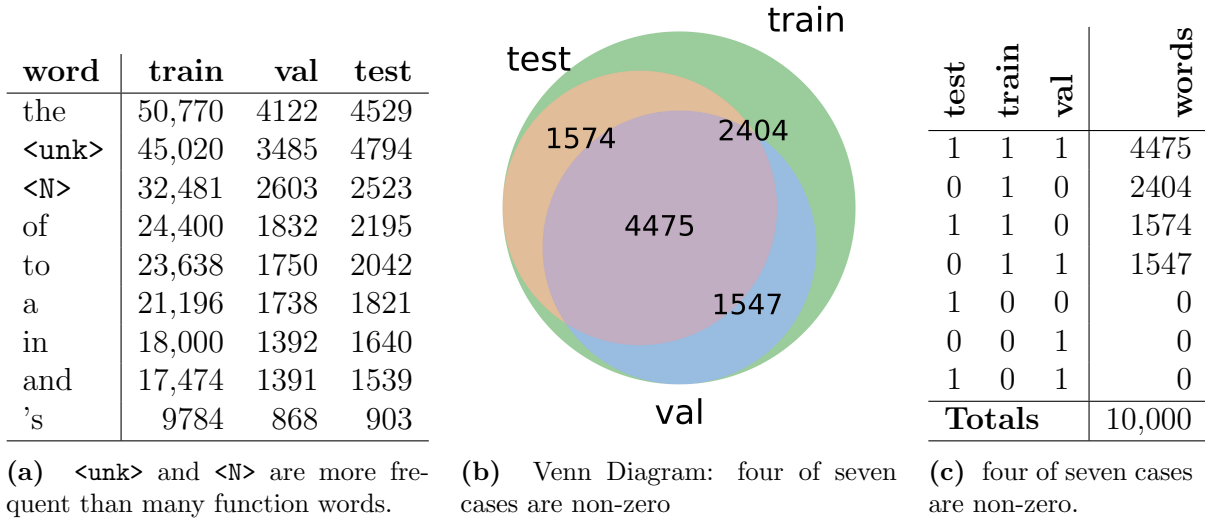


Figure 2. When PTB was repurposed to study perplexity, the vocabulary was reduced to 10,000 words by assigning rare words to <unk> and numbers to <N>. This construction makes it (too) easy to predict the next “word,” especially because the test set does not generalize beyond the training set. That is, there are no words in the test set that do not appear in the training set.

meaningful) the task becomes. Issues with this processing have been called out elsewhere such as this discussion of WikiText-103 on Papers with Code (PwC):¹⁷

Compared to the preprocessed version of Penn Treebank (PTB), WikiText-2 is over 2 times larger and WikiText-103 is over 110 times larger. The WikiText dataset also features a far larger vocabulary and retains the original case, punctuation and numbers – all of which are removed in PTB. As it is composed of full articles, the dataset is well suited for models that can take advantage of long term dependencies.

Venn diagrams in Figures 2b and 2c point out an additional undesirable consequence of this processing. Normally, we would expect to see non-zero counts in all seven cells of the Venn diagram, but so many rare words have been removed that many of the cells are empty. In particular, there are no words in the test set that were not seen during training, casting serious doubts on the ability of this benchmark to measure generalizations beyond the training set.

We, as a community, should be more careful with credit assignment, especially when introducing serious problems into other people’s work. It is ironic that the community refers to the text-only version of the Penn TreeBank as “PTB” since this version has little to do with P (Penn) or T (trees). As explained in the LDC version in footnote 16, the text was collected elsewhere, largely from the 1989 Wall Street Journal (WSJ) and the 1961 Brown Corpus. Penn contributed the trees, but not the text (or the inappropriate processing).

¹⁷<https://paperswithcode.com/dataset/wikitext-103>

6 Common Assumptions in Estimates of Perplexity

There are many potential pitfalls for benchmarks for estimating perplexity. In addition to inappropriate processing, there can also be leakage and deviations from i.i.d. assumptions, as will be discussed below. Before those discussions, let us review methods for estimating perplexity.

Suppose we have a training set and a test set. The training set is used to fit $\hat{Pr}(w|history)$, estimates of the probability of the next word (or token) given the previous context (*history*). At test time, we apply this model, $\hat{Pr}(w|history)$, to estimate probabilities based on a sample of strings (the test set). Finally, we use a geometric mean to aggregate the probabilities to produce a single number.

Theoretically, entropy, H , is the number of bits required, on average, to predict the next symbol. Perplexity is simply 2^H . Perplexity can be interpreted as a branching factor in a search space, or the typical number of choices.

Entropy is defined in terms of an (unknowable) probability distribution. If we have a set of events $\{e_1, e_2, \dots, e_n\}$ with probabilities $Pr(e_1), Pr(e_2), \dots, Pr(e_n)$, then $H = -\sum_i Pr(e_i) \log_2 Pr(e_i)$.

In practice, since we do not know the probabilities, we use the training set to fit estimates, $\hat{Pr}$. And then, we treat the test set as n samples, s_i , from the event space. As mentioned above, perplexity can be thought of as the branching factor in a search space. That is, the number of choices for sample s_i is $\frac{1}{\hat{Pr}(s_i)}$. If we have n samples, then the “typical” number of choices (perplexity) is:

$$2^{\hat{H}} = \left(\prod_{s=1}^n \frac{1}{\hat{Pr}(s)} \right)^{1/n}$$

The geometric mean can be rewritten as an arithmetic mean of logs:

$$\hat{H} = -\frac{1}{n} \sum_{i=1}^n \log_2 \hat{Pr}(s)$$

In general, $\hat{H} \geq H$, where H is based on unknowable probabilities, Pr , and $\hat{H}$ is based on estimates, $\hat{Pr}$. We refer to $\hat{H}$ as cross entropy, and H as entropy. The gap between $\hat{H}$ and H becomes smaller and smaller with better and better estimates of Pr .

This approach assumes that the test set consists of i.i.d. samples, s_i , drawn from a log normal distribution. Obviously, the test set should be different from the training set. If there is too much leakage, then the test set will fail to measure generalizations to unseen inputs. In practice, there are a number of challenging realities. Real documents have all sorts of non-i.i.d. structures such as topics, genre, style, discourse structure, plots, character arcs, languages and more. There is a literature on robust statistics where it is recommended that one split the test set up into many subsets and estimate many means (and moments) over many subsets. In this way, one can identify outliers, subsets of the test set that are surprisingly easy and surprisingly hard. There are often interesting explanations for many of the outliers as will be discussed below. These explanations typically involve deviations from i.i.d. assumptions. As the cliché goes, all models are wrong, but some are useful.¹⁸ We know that realistic texts are not i.i.d., but we want to know which are good enough, and which are truly problematic.

¹⁸https://en.wikipedia.org/wiki/All_models_are_wrong

The discussion of Wikitext-103 above suggested that bigger test sets are better. In fact, that may not be the case. Under the log normal assumption mentioned above, the size requirements on the test set involve the variance. If we split the test set up into many subsets and they all produce similar results as expected under a log normal distribution, then there is little point to increasing the size of the test set. Unfortunately, the literature rarely reports results on subsets of the test set, let alone comparisons of such results with expectations under a log normal distribution.

Another concern is coverage. The test set should be a representative sample of a population of interest such as the text that the application is likely to see in the future. Test sets are often based on news wire and Wikipedia, but it should be possible to construct more representative samples. The problem with the `<unk>`-prediction task above is that real use cases are unlikely to contain `<unk>` tokens.

6.1 Representative Samples: Leakage, Duplication, Balance and Age

Estimating perplexity has become more challenging now that LLMs are trained and tested on data from undisclosed sources. When corpora were curated by dictionary publishers, they were more careful about quality, especially clearing rights and documenting provenance.¹⁹ These days, the research community is less careful. LLM companies consider their sources to be trade secret.

Given these realities, duplication may be unavoidable. There is considerable literature on duplication in discussions of web crawling [Manasse, 2015], but less so in discussions of perplexity and LLMs. As a result, it is likely that LLMs have been trained on the test data, or some data that is (too) similar to the test data. Lee et al. [2022], Carlini et al. [2023] and Nasr et al. [2023] have been studying duplication in standard corpora that are likely to be used to train chatbots. They have been using hashing methods [Broder, 1997] and suffix arrays [Manber and Myers, 1993] to estimate memorization in chatbots. When bots are prompted with the beginning of long sequences such as 50-grams that are frequent in corpora that were probably used to train the bot, many bots often return the rest of the repeated sequence. Larger bots with more parameters are more likely to memorize more 50-grams with smaller frequencies. Estimates of perplexity are based on strong i.i.d. assumptions that are violated by this kind of duplication and leakage. A corpus should be a representative sample of what the model is likely to encounter at inference time.

The literature on corpus-based methods has a long history. There used to be more concern for duplication and balance. Corpora such as the Brown Corpus were designed to be a representative sample of Contemporary American English. When they collected this sample in the 1960s, they stratified the sample over different genre. Fiction is different from non-fiction, and local news is different from international news. For example, the word “said” is as frequent as a function word in the news, but extremely rare in technical abstracts. Estimates of perplexity based on the Brown Corpus will be relatively high compared to the Wall Street Journal because the WSJ represents a relatively homogeneous genre unlike the Brown Corpus, which contains, by construction, a diverse variety of genres.

In addition to balance, we should be concerned about age. The use of the phrase “present-day” in descriptions of the Brown Corpus²⁰ do not age well since the Brown Corpus was based

¹⁹<https://www.nytimes.com/2023/12/27/business/media/new-york-times-open-ai-microsoft-lawsuit.html>

²⁰https://en.wikipedia.org/wiki/Brown_Corpus

on works published in the United States in 1961. The WSJ data in PTB was from 1989; that is also far from “present-day” though not as dated as the Brown Corpus. Dictionaries are updated frequently because it is well-known that languages evolve considerably over decades.

6.2 A Constructive Proposal for Estimating Perplexity

The discussion of WikiText-103 above called out size and processing, but not duplication, leakage, sampling and age. Is size an advantage or a disadvantage? When the test set becomes larger and larger, it becomes harder and harder to know where it came from, or what it represents. It also becomes harder to know if there is leakage. Who knows what commercial bots are trained on, but it is hard to imagine that they have not been trained on Wikitext-103.

A suggestion in [Borgeaud et al., 2022] is to estimate perplexity on text from the “future.” In their Appendix A2, they describe a test set of Wikipedia articles that were created after training time. In addition to Wikipedia, many sources of text have a natural time dimension. Two examples are academic papers and news articles, both of which have publication dates. Suppose we train on publications up to time t , and test at time $t + h$, where h is the forecasting horizon. We would expect estimates of perplexity to improve with t and degrade with h .

In addition to publishing a single number, estimates of perplexity should also provide details. Perplexity is a geometric mean. One would hope that the values that contribute to the mean are distributed as log normal. How appropriate is that assumption? A standard method for testing that assumption is `qqnorm`,²¹ a special case of a q-q (quantile-quantile) plot.²² This plot will help us find outliers from assumptions of normality. After identifying outliers, we should decide if they are a concern.

It has been our experience that size is not always an advantage. Larger collections tend to have more corner cases (outliers). Since outliers can have a large impact on estimates of perplexity, one should check to see how sensitive estimates are to particular outliers. If the estimate depends on just a few outliers, we should decide how much we trust those outliers. If the outliers are uninteresting corner cases, then the estimate of perplexity is less trustworthy than if the outliers are central to the population of interest.

Large collections of news will include many outliers that can lead to iffy estimates of perplexity. Examples include stories with both high and low estimates of perplexity:

- Surprisingly low perplexity: file stories, tables, weather, plagiarism, duplication.
- Surprisingly high perplexity: documents in languages other than English, summaries of the news of the day/week/year.

By file stories, we are referring to stories that appear on slow news days, especially when much of the staff is on vacation. Tables on votes in congress are often predictable. The list of representatives does not change much, and they usually vote along party lines. The main difference from one vote to the next is the question they are voting on, but that is a relatively small fraction of the story.

At the other end of the spectrum, we often find a few documents in languages that we were not expecting. One might expect papers in Semantic Scholar to be in English, but 17% are in other languages. One might also expect the beginning and the end of a news article to be on the same

²¹<https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/qqnorm>

²²https://en.wikipedia.org/wiki/Q%E2%80%93Q_plot

topic, but there are many summary articles that combine a couple of sentences from a couple of different articles.

It becomes much easier for the community to become aware of corner cases such as these if we not only publish estimates of perplexity on standard collections, but we also share

- estimates of perplexity on articles from the future, such as papers from Academic Search websites, grouped by publication year,
- as well as details that make it easy to identify outliers and test assumptions such as log normal distributions.

7 Labeling and Inter-Annotator Agreement

Most of the AI-based solutions require ground truth data that is created in a labeling step. Large-scale labeling (annotations) is very hard; it needs software development to get good results, and requires quality control processes and metrics [Alonso, 2019; Garcia-Molina et al., 2016; Roh et al., 2021]. Many aggregation models have been proposed for simple annotation tasks, but far less work has considered more complex tasks that involve open-ended, multivariate, or structured responses [Braylan et al., 2023].

There is very recent work on using LLMs for labeling as a mechanism to scale up the process and reduce the human cost. Using humans for labeling is expensive and LLMs offer advantages in terms of cost and performance. That said, the same principles with respect to reliability and quality control apply here. Faggioli et al. [2023] provides a spectrum of human-machine collaboration for different configurations. Thomas et al. [2023] compare LLM labels with TREC labels and report variations on κ for inter-annotator agreement.

Data wrangling, task design, task complexity, and aggregation techniques are still required. This is hard work that needs to be done for labeling. As previous work has noted, in practice nobody wants to work on data issues, and everyone expects clean labels. Labeling keeps getting harder as more complex tasks require more labels.

Unreliability results from two factors: (1) disagreements among independent ways of generating data about the phenomena in question and (2) lack of information (variation) in the resulting data to answer given research questions about these phenomena. In content analysis [Krippendorff, 2018], the first is assessed by agreement coefficients, such as α , applied to how several observers unitize the same continuum and/or record, describe, or code the same set of units. The second becomes evident by tracing the information in data throughout an analytical process to its results. When variation is absent, data cannot be correlated with anything; and when undetected noise in the data is not eliminated during an analysis, the answers to research questions may well be unrelated to the phenomena of interest.

8 Beyond Unambiguous Uncontroversial Factoid Queries

Many benchmarks assume there is a single correct answer in the answer key, but many tasks are not so simple: translation, web search, etc. It is standard practice for translation and web search to score systems with metrics such as BLEU and NDCG that aggregate over multiple references

	Fact	Opinion
Entity	On which day will iphone be released?	Top 10 movies of 2017
Description	Why are firetrucks red?	How good is the Toyota Corolla?
Yes/No	Is 39.5 degrees a high fever?	Does learning to play Go improve intelligence?

(a) Six Types of Questions in DuReader (Table 2 of [He et al., 2018])



(b) Candidate labels: *baseball cap, cap, green hat, hat* and *head*; which label is in the answer key?

Figure 3. The most interesting questions are rarely objective and unambiguous.

(answers in the answer key). Question-answering benchmarks such as SQuAD and TREC-8 focus on factoid questions, but some of the most interesting questions involve subjective opinions, as illustrated by the following three examples:

1. DuReader, for example, contains six types of questions extracted from Baidu query logs (in Chinese). Table 3a shows English glosses for the six types which include opinions in addition to factoids.
2. Historically, vision tasks have tended to label bounding boxes, but Figure 3b shows considerable room for annotators to disagree with one another.
3. Progress on part of speech tagging was limited by benchmarks that left too little room for differences of opinion. One might think that the difference between a noun and a verb would be unambiguous, but it turns out that annotators disagree about 3% of the time. When a machine differs from an annotator, the machine is wrong. But when two judges disagree, it is a difference of agreement. We could make progress on tagging if we could distinguish machine errors from differences of opinion. It would help to enhance answer keys with multiple answers, and replace metrics like accuracy with metrics like BLEU and NDCG that aggregate over multiple references.

In more recent work [Mohamed et al., 2024], we have been trying to move past unambiguous objective labels by asking annotators to add emotion labels and captions in 28 languages to WikiArt.²³ For a particular portrait, many annotators thought the woman was beautiful but some objected to her dress as too revealing. One reviewer insisted on an ethics review, objecting to the objection to the dress. We object to the objection to the objection; it is inappropriate to impose our views on the rest of the world. We are not siding with one annotator over another, but rather than remove bias from corpora, we hope to build models that will be more aware of regional sensitivities to topics such as: dress, religion and alcohol. In addition to publishing a benchmark, we offer baseline implementations of captioning systems. Hopefully, the community will accept the challenge and come up with even better systems that embrace diversity over many regions, cultures and languages.

²³<https://www.wikiart.org/>

Web, news and social media offer many different perspectives and points of view. Search engines have developed strategies for addressing these realities, but QA systems and chatbots tend to focus on an over-simplified view of the world with “right” and “wrong” answers, based on a single “possible world.” American bots are trained on American corpora; these bots currently lack a historian’s ability to approach conflicts from multiple perspectives.

The second author asked his NLP class to use bots to help them write essays about a controversial topic (the Opium Wars) from multiple perspectives including both the East and the West [Church, 2024]. He had hoped that international students would rewrite the output from the bot to reflect diverse perspectives, but most students found it easier to hand in the output from the American bot, as is. American bots do not mention “the century of humiliation,”²⁴ a view that is motivating efforts to compete with the West in artificial intelligence so China does not fall behind in technology like it did during the Opium Wars, as suggested by some historians [Platt, 2019].

When bots and QA systems over-simplify the truth and take our side of conflicts, that could create dangerous filter bubbles, causing trade wars and worse. More generally, a major challenge for AI (and evaluation of AI) is to move beyond unambiguous, uncontroversial factoid questions. Many questions do not have simple right and wrong answers, but rather lots of opinions, sensitivities, and perspectives. Typical workloads on the web, news and social media go well beyond unambiguous uncontroversial factoid queries.

9 Conclusions: List of Principles

We propose to think about evaluation benchmarks the same way the community develops software techniques. In short, it needs to address a real customer problem, the workload is representative of the real-world, the metric is easy to interpret, and the data set evolves over time.

There are a few questions that the benchmark should focus on. How realistic is the task? Are we trying to solve a common use case or is this something that looks a bit artificial? Commercial relevance?

We outline the following list of principles that we think need to be taken into consideration when creating a new evaluation benchmark. At the same time, depending on the stage of the development process, some of these principles might take priority.

1. Realistic workloads: characterize the intended use case and audience.
2. Validity. Is the evaluation task relevant to the intended use case? What are we supposed to learn from the benchmark? A benchmark should make a point (and not too many points).
3. Reliability with respect to the data.
 - Inter-annotator agreement scores and disagreement analysis.
 - Labeling and annotation. How was it done? Availability of task instructions.
 - Outlier analysis and robust statistics: can we trust the metrics?
 - Simplicity? A benchmark should be large enough to make the point (and no larger).

²⁴<https://www.uscc.gov/sites/default/files/3.10.11Kaufman.pdf>

4. Maintenance.

- Include a feedback loop mechanism to maximize adoption.
- Workload evolution.
- Lessons learned and addendum(s).
- Lifecycle and deprecation.

5. High standards (for high-stakes use cases).

- Use of established software engineering and data management techniques (e.g., code review, versioning, configurations, dependencies, and testing).
- How the data was sourced? Can the data set be generated again easily?
- Clean and well-documented data model.

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